Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with 2+ years building scalable **Python** backend services and production **generative AI** systems. Experienced in **distributed systems**, asynchronous **LLM** job orchestration, **Azure**/**AWS** cloud platforms, unit and integration testing, and reliable high-throughput AI workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Advanced NLP, Database Systems, Computer Networks

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, Operating Systems, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built performant backend services with **Python**/FastAPI and **Azure**, architecting fault-tolerant async **LLM** orchestration that processed **1K+ AI jobs** with explicit job-state transitions, failure isolation, and **3x** throughput gains under concurrent load.
- Developed scalable **PostgreSQL** data models and integration pipelines processing **5K+ monthly jobs** at **99%+ reliability**, writing validation checks and participating in code reviews while troubleshooting production issues across cloud-hosted systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Delivered production **generative AI** services integrating **LLM APIs** (GPT-4o, Claude, Qwen-VL) and OCR into backend workflows, serving **10K+ daily requests** with modular **Python**/Flask services and **MySQL**/AWS RDS persistence.
- Implemented unit and integration tests with **GitHub Actions CI/CD**, pair-programming with senior engineers on design decisions and reducing manual turnaround by **60%** while maintaining stable API performance under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built distributed processing pipelines over **100K+ trajectory logs**, applying transformer-based modeling and evaluation workflows that improved long-horizon planning stability by **15%** through rigorous experimentation and clean backend design.
- Collaborated on research-to-production tooling with structured logging and reproducible pipelines, sharing implementation knowledge across teammates while optimizing data preparation and system integration for large-scale AI workloads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **Python** microservices and **AWS** ETL pipelines with **20+ PyTest** unit and integration suites, improving distributed API response times by **30%** and Jenkins pipeline efficiency by **68%** across cross-platform environments.
- Troubleshot system integration defects and recommended process improvements during Agile code reviews with Microsoft/GitHub partners, producing documentation and reusable backend modules for reliable large-scale data-sharing services.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed backend services across an **8-service microservices** platform with **Docker** and Jenkins **DevOps** pipelines, supporting high-throughput operational workflows syncing **2M+ records** with **95%+** successful build validation.
- Contributed to cloud-native deployment automation and quality assurance practices, cutting release cycles by **50%** while partnering on troubleshooting, system integration, and performance tuning across distributed application layers.

Skills

Backend & AI: Python, C# (familiar), FastAPI, Flask, backend services, generative AI, LLM APIs, async job orchestration, distributed systems, scalability, REST APIs, microservices, event-driven architecture (familiar), domain-driven design (familiar)
Cloud & Quality: Azure, AWS, Kubernetes (familiar), serverless (familiar), cloud-native, Docker, GitHub Actions, CI/CD, Jenkins, PostgreSQL, MySQL, SQL Server, PyTest, unit testing, integration testing, Git, Agile, troubleshooting, AI-assisted development

Projects

Linguistix | *Python, AWS ECS, Docker, CI/CD, AI APIs*

- Deployed scalable **Python** backend services on **AWS ECS** with CodePipeline **CI/CD**, supporting AI-ready ingestion workflows and reliable releases across development, staging, and production environments.

FinNexus Live | *TypeScript, Kubernetes, High-Throughput Backend*

- Built a cloud-native fintech platform with **Kubernetes** deployments and **CI/CD** automation, handling **1M+ transactions** with secure APIs and real-time updates at production scale.

CaptionCraft Live | *Generative AI, MongoDB, SaaS Backend*

- Developed a generative AI SaaS backend using transformer models and **MongoDB**, processing **1K+ assets** with scalable storage, automated workflows, and production-ready API integrations.